

Does the J-space trade off against written chain of thought as problems get harder?

Homework Zero — Fall 2026. Model: Qwen3-4B. Code: <https://github.com/avchadha/AI-Alignment>.

1. Hypothesis

Recorded and committed before any main runs (report/hypothesis.md; see git history): **ablating the J-space impairs the multi-hop reasoning needed to solve mathematics problems; the impairment grows with problem difficulty; and explicit chain of thought (CoT) counteracts it, increasingly with the amount externalized.** Predictions: (1) direct answers suffer more from J-space ablation than CoT answers, and far more than from a matched-norm random ablation; (2) the control-ablated gap shrinks monotonically with thinking budget; (3) relative loss grows with difficulty (GSM8K → MATH-500 → AIME; MATH-500 levels 1-5); (4) ablated models write longer CoT when permitted.

2. Experiment design

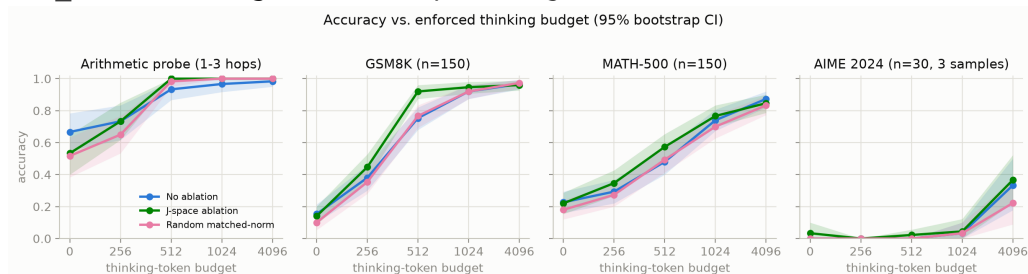
Three ablation arms × five thinking budgets on fixed problem sets (paired comparisons). **Arms:** control; J-space ablation — at every token position across the workspace layer band, remove the residual stream's projection onto the span of the $k = 10$ most strongly activated J-lens vectors, never ablating a clean pass's top-10 tokens (the paper's procedure); random matched-norm — k fixed random directions in the same band, rescaled per position to remove exactly the norm the J-space ablation would remove; this arm separates J-space-specific damage from generic degradation. **CoT axis:** budgets {0, 256, 512, 1024, 4096} thinking tokens (Qwen3's native thinking mode), enforced by the harness: over-budget generations receive a forced `</think>`, and every arm's answer segment is teacher-forced to The final answer is `\boxed{...}` (48-token cap; instruction-only direct-answer conditions leaked CoT in the pilot). **Datasets:** GSM8K (openai/gsm8k test, $n = 150$, seeded), MATH-500 (HuggingFaceH4/MATH-500, $n = 150$, 30 per level), **AIME 2024** — HuggingFaceH4/aime_2024, the 30 problems of the 2024 AIME I and II exams, 3 samples per problem — and a synthetic 1/2/3-hop arithmetic probe (60 seeded problems) added after calibration revealed a direct-answer floor (§3). Each opposite outcome is visible by construction: no rescue leaves ablated curves flat; generic damage makes the random arm track the J-space arm; a reversed difficulty interaction shows relative loss falling. **Metrics:** accuracy with 95% bootstrap confidence intervals (AIME samples averaged within problem), paired McNemar tests, retained accuracy, CoT length, and clip and extraction-failure rates.

3. Experimental details

The intervention uses a pre-fitted community Jacobian lens for Qwen3-4B (neuronpedia/jacobian-lens, wikitext, official recipe), accepted after it reproduced the paper's walkthrough readout. The ablation is implemented here (src/jspace/ablation.py; the official repository has readout code only): J-lens vectors fold the final RMSNorm into the unembedding, the joint projection onto the selected span is removed by QR, and the clean-pass exclusion runs a second hook-free pass ($\approx 2\times$ cost). Ranking uses raw lens logits (cosine ranking was calibrated and agreed). On 30 held-out calibration problems, control direct-answer accuracy is 16.7% versus 86.7% with 1,024 thinking tokens; the medium band (layers 14–29 of 36) was frozen for the largest direct-answer drop (to 6.7%; tied with heavy under logit) with SST-2 intact under every band and rule (89–90%, vs 89% control). The J-space-versus-random comparison was excluded from calibration criteria. Main run: 6,750 generations (3 arms × 5 budgets × 450 items), one H100, ≈ 13.5 GPU-hours; judging by balanced-brace `\boxed{...}` extraction with math-verify equivalence for MATH-500. Reproduction: README.md, configs/frozen.yaml.

4. Experimental results

All 6,750 generations completed; no extraction failures. Full tables: results/analysis_main.txt, results/extra_stats.txt; figures 1–5: report/figs/.



budget	GSM8K ctrl	GSM8K J-abl	GSM8K rand	MATH ctrl	MATH J-abl	MATH rand
0	.153	.140	.100	.227	.220	.180
256	.380	.447	.353	.293	.347	.273
512	.753	.920	.767	.480	.573	.493
1024	.920	.947	.920	.740	.767	.700
4096	.967	.960	.973	.873	.847	.833

(a) No selective impairment at budget 0 (prediction 1 not supported). GSM8K/MATH-500 direct answering sits at the predicted floor; J-ablation moves it ~ 1 point (McNemar $p = 0.80 / 1.00$). On the arithmetic probe, whose baseline has headroom (66.7%), J-ablation does impair direct answering (53.3%; $p = 0.0078$) with a clean hop gradient — 1-hop unaffected, 2-hop 75% $\rightarrow$ 55%, 3-hop 25% $\rightarrow$ 5% — but the random arm lands in the same place (51.7%; $p = 1.0$): multi-hop-specific damage, not J-space-specific.

(b) A J-space-specific gain at intermediate budgets. At budget 512 the J-ablated arm exceeds control by 16.7 points on GSM8K (92.0% vs 75.3%; McNemar 27 vs 2, $p < 10^{-5}$) and 9.3 points on MATH-500 (57.3% vs 48.0%; $p = 0.0043$), with the same sign at 256. Unlike (a), this is specific to the J-space directions: the random arm tracks control (76.7% / 49.3% at 512), and paired J-space-versus-random tests give $p < 10^{-4}$ (GSM8K, 512), $p = 0.024$ (GSM8K, 256), $p = 0.017$ (MATH-500, 512).

(c) The mechanism is shorter CoT (prediction 4 inverted). At the effectively unconstrained 4,096 budget, J-ablated chains are shorter: 1,202 versus 1,703 mean thinking tokens on GSM8K (random: 1,642), 853 vs 1,049 on the probe, 2,326 vs 2,657 on MATH-500. The J-ablated arm is clipped less wherever clipping is common (GSM8K at 1024: 41% vs 69%), and the gain in (b) concentrates where control chains are truncated mid-reasoning.

(d) Dose-response (prediction 2, weak form). All arms rise steeply and converge by budgets 1024-4096 (GSM8K 96-97%; MATH-500 83-87%). CoT fully rescues the arithmetic impairment (J-ablated 100% at budget 512; control 93.3%, random 98.3%) — but from damage that was never J-space-specific.

(e) No predicted difficulty interaction (prediction 3 not supported). Within MATH-500 at budget 512 the J-ablation advantage appears at levels 1-4 (+.10, +.17, +.10, +.10) and vanishes at level 5 (.23 both). AIME floors below 4096 (0-4%) with no significant differences at 4096 (33.3% / 36.7% / 22.2%); 89-94% of chains still clip there, so that condition largely measures truncation.

5. Analysis of results

Only the weak form of prediction 2 (rescue by CoT, with dose-response) survives: prediction 1's impairment appears only off the floor and is matched by random perturbation; prediction 3's interaction is absent; prediction 4 inverted. The one robustly J-space-specific effect is opposite in direction to the hypothesis: removing the top-k J-lens directions makes written CoT roughly 25% shorter at equal or better accuracy. A post-hoc transcript probe (scripts/verify_reflection.py; 20 GSM8K problems regenerated with full think-text) shows the omitted content is disproportionately reflective: J-ablated chains carry fewer reflection markers per thinking token (6.7 vs 9.9 per 1,000; "wait" 1.9 vs 3.2) — roughly half the double-checking and re-derivation loops per problem. This suggests the ablated directions carry self-monitoring or verbalization content rather than load-bearing intermediate state: they modulate how much the model narrates rather than serving as an internal scratchpad. Qualifications: floor effects (with externalization prevented, un-ablated Qwen3-4B scores 15-23% on GSM8K/MATH-500 and 0% on AIME — little internal reasoning to ablate, cautioning against extrapolation to small thinking-tuned models) and alternatives: the wikitext-fitted community lens may capture a shallower subspace, the random arm matches removed norm but not its residual-basis location, and sampling adds variance. None of these plausibly explains the intermediate-budget gain, which replicates across two datasets with the random arm flat. Next: a paper-faithful lens fit; k and band sweeps with delta-norm audits; clamping single J-vectors to isolate the shortening direction; faithfulness checks on shortened chains; a larger model with an off-floor direct baseline.